Gursimran Singh

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Seeking Summer 2027 research internships · Expected graduation May 2028

Research Summary

Ph.D. researcher in programmable network systems and security, with experience implementing and evaluating P4 prototypes on Intel Tofino hardware and developing Internet-measurement tools. Research interests include stateful network security, reliable in-network computation, and Internet resilience.

Education

Oklahoma State University — **Jan 2026 – Present**
Ph.D. in Computer Science · Advisor: Dr. H. B. Acharya — *Expected May 2028*

Rochester Institute of Technology — **2021 – 2025**
Ph.D. coursework, Computing and Information Sciences · GPA 3.80/4.0 · transferred to OSU with advisor — *Rochester, NY*

University at Buffalo (SUNY) — **2019 – 2021**
M.S. in Computer Science and Engineering · GPA 3.52/4.0 — *Buffalo, NY*

University of Delhi, Sri Guru Gobind Singh College of Commerce — **2016 – 2019**
B.Sc. (Hons.) Computer Science — *New Delhi, India*

Publications

[P1] **Gursimran Singh**, H. B. Acharya, and Minseok Kwon. **Programmable Data Planes for Network Security.** In *Proc. 17th International Conference on Networks & Communications (NeTCoM 2025)*, Computer Science & Information Technology (CS & IT) vol. 15, no. 13, pp. 163–180, July 2025. DOI: 10.5121/csit.2025.151314 · arXiv:2507.22165.
Systematization of security applications on P4 programmable switches, their hardware constraints, and the architectural workarounds that make them feasible.

[P2] **Gursimran Singh** and H. B. Acharya. **POSTER: A Cyberspace Study of the Russia–Ukraine War.** In *Proc. ACM ASIA Conference on Computer and Communications Security (ASIA CCS 2023)*, pp. 1016–1018, July 2023. DOI: 10.1145/3579856.3592822.
Poster paper: measurements of website access from Ukraine, Russia, and the United States collected in March and July 2022, with a soft-failure classifier and a multi-protocol traceroute.

Additional first-author manuscripts are currently under submission (details available upon request).

Research Experience

Doctoral Researcher, advised by Dr. H. B. Acharya — **2021 – Present**
Oklahoma State University (2026 – present) · Rochester Institute of Technology (2021 – 2025)

Programmable data-plane security on Intel Tofino — 2021 – Present

- Implemented packet-inspection and stateful-filtering prototypes in P4 on Intel Tofino/Tofino2, working within parser, stage, memory, and instruction-set constraints.
- Developed and evaluated packet-processing logic using the P4 Studio SDE and BF-RT control plane.
- Investigate the relationship between completed inspection, enforcement timing, and throughput under constrained hardware resources.
- First-authored a systematization of programmable-data-plane security research, published at NeTCoM 2025 [P1].

Reliable in-network computation (ongoing research) — 2026 – Present

- Investigating which computation belongs in a programmable switch, on a SmartNIC/DPU, or on the host (CPU/GPU) for performance and robustness, and how in-network computations can recover from device failures while preserving correct results across that heterogeneous hardware, and at what cost.

Internet measurement, censorship, and resilience — 2022 – Present

- Designed and implemented measurement tooling for the Russia–Ukraine study: a classifier for *soft* failures of website access and a multi-protocol traceroute used to investigate where filtering responses may originate.
- Collected and analyzed measurements from Ukraine, Russia, and the United States in March and July 2022 (ACM ASIA CCS 2023 [P2]).
- Ongoing: how reachability and paths change during disruption, and how measurement uncertainty and probe behaviour affect the conclusions.

Industry Experience

Keysight Technologies **Sep 2026 – Present**

R&D Software Engineering Intern (FPGA development for AI infrastructure testing solutions) · part-time, remote

- Contribute software components and prototyping for FPGA-based AI infrastructure testing solutions, collaborating with hardware engineers.

Teaching Experience

Adjunct Instructor, Rochester Institute of Technology **2022, 2024 – 2025**

CSEC-140 Intro to Cybersecurity (2022, 2025) · CSCI-455 Principles of Cybersecurity (2024, 2025) · GCIS-123 Software Dev. & Problem Solving I (2025)

Teaching Assistant, Rochester Institute of Technology **2023 – 2024**

CSCI-243 Mechanics of Programming · CSEC-759 Advanced Networking & Distributed Systems Security (graduate seminar)

Teaching Assistant, Oklahoma State University **Spring 2026**

Distributed and Cloud Computing

Research & Capstone Mentor **2022 – 2025**

Mentored three master's students and two undergraduate students on research and capstone projects

Professional Service

Artifact Evaluation Committee, EuroSys 2027 (both submission cycles) 2026 – 2027
Artifact Evaluation Committee, ACM PPoPP 2027 2026 – 2027
Junior Program Committee, ACM ICS 2027 (International Conference on Supercomputing) 2026 – 2027
Technical Program Committee, ACM S3 2026 student workshop (co-located with ACM MobiCom 2026) 2026
Shadow Technical Program Committee, ACM CoNEXT 2026 2026

Leadership

President, Computer Science Graduate Student Association (CS-GSA), Oklahoma State University: lead a seven-officer board; orientation, professional-development talks, networking, travel/funding advocacy 2026 – 2027
Council Member, Doctoral Student Association, Rochester Institute of Technology 2025
Vice-President, CSE Graduate Student Association (CSEGSA), University at Buffalo 2020 – 2021
President (2018 – 2019) and **Vice-President** (2017 – 2018), ACE (Computer Science Society), SGGSCC, University of Delhi; organizer, CYTHON-18 tech fest 2017 – 2019

Competitions & Honors

MITRE Embedded CTF (eCTF) 2021, University at Buffalo team: implemented RSA key-generation and primality-testing routines in C for an embedded-security competition prototype 2021
3rd place, 2020 regional round (hosted by RIT's ESL Global Cybersecurity Institute), Collegiate Penetration Testing Competition (CPTC), University at Buffalo team Nov 2020
"Stars of the College" profile feature, SGGSCC Annual Report 2018

Technical Skills

Languages: C, C++, $P4_{16}$, Python, Java, Bash; CUDA and MATLAB (project experience)

Programmable networking (research use): Intel Tofino / Tofino2 (TNA), P4 Studio SDE, BF-RT control plane, parsing/deparsing, recirculation, mirroring, match-action pipeline design

Exposure only: BMv2, eBPF/XDP, SmartNIC/DPU and FPGA data paths

Measurement & security tooling: Scapy, Wireshark, Zmap, Nmap, traceroute-based path measurement, Aircrack-ng, Kali Linux, IDA, OllyDbg

Systems & data: Linux, Git, VirtualBox lab networks, Pintos kernel development; scikit-learn, Keras, NumPy, Pandas

Certifications & Training

Intel Connectivity Academy: ICA-XFG-101 ($P4_{16}$, Tofino family, P4 Studio SDE); ICA-XFG-201 – 206 (Advanced TNA: parsing/deparsing, recirculation, mirroring) 2023 – 2024
OffSec Wireless Professional (OSWP) and **Kali Linux Certified Professional (KLCP)**, Offensive Security 2023

Selected Projects

Core OS components in Pintos (C): kernel-level threading and the system-call interface.

Task offloading for QoS with a dynamic ANN (C, C++, CUDA): run-time adaptation to query parameters (max hops, query volume).

Reliable transport protocols and multi-client TCP chat (C/C++): Alternating-Bit, Go-Back-N, and Selective-Repeat over an unreliable channel.